Chapter 11 Notes

Section 0

Notes

Note that elementarily equivalence only requires pure formulas to be equivalent for the following reason: $A_1 \neq A_2$ may be true, and if it is, then if $a \in A_1$ and $a \notin A_2$, then there are formulas either true or false in (A_1, R_1) (formulas involving a as a constant) where truth in (A_2, R_2) isn't even defined.

Exercise 0.1

Suppose π is an isomoprhism from (A_1, R_1) to (A_2, R_2). We show by complete induction on the degrees of formulas that for any pure formula $\phi(x_1, ..., x_n)$ and any elements $a_1, ..., a_n$ of A_1, the sentence $\phi(a_1, ..., a_n)$ is true in (A_1, R_1) iff $\phi(\pi(a_1), ..., \pi(a_n))$ is true in (A_2, R_2), and, for any pure sentence ϕ, ϕ is true in (A_1, R_1) iff ϕ is true in (A_2, R_2).

In the base case, Pab (for elements a, b in A_1) is true in (A_1, R_1) iff aR_1b iff $\pi(a)R_2\pi(b)$ (because π is an isomorphism) iff $P\pi(a)\pi(b)$ is true in (A_2, R_2). Thus the base case holds.

Now suppose that the result holds for all formulas of degree less than n. That is, for any pure formula $\phi(x_1, ..., x_m)$ (of a degree less than n) and any elements $a_1, ..., a_m$ of A_1, $\phi(a_1, ..., a_n)$ is true in (A_1, R_1) iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2), and for any pure sentence ϕ, ϕ is true in (A_1, R_1) iff ϕ is true in (A_2, R_2). On the one hand, we consider a pure formula with free variables of degree n. That is, we consider some pure formula $\phi(x_1, ..., x_m)$ and elements $a_1, ..., a_m$ of A_1. There are then multiple cases.

In the first case, $\phi(x_1, ..., x_m) = (\psi \wedge \gamma)$ for some pure formulas ψ and γ of degrees less than n. Either (1) neither ψ nor γ are sentences; (2) ψ is a sentence and γ is not; (3) ψ is not a sentence and γ is a sentence. We consider the first case - the other two are nearly identical. In that case, because neither ψ nor γ are sentences, $\psi = \psi(y_1, ..., y_k)$ for some $k < n$. Similarly $\gamma = \gamma(z_1, ..., z_l)$ for some $l < n$. Then $\phi(x_1, ..., x_m) = (\psi(y_1, ..., y_k) \wedge \gamma(z_1, ..., z_l))$. Then $\phi(a_1, ..., a_m) = (\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ where each $b_i = a_j$ for some j (and each $c_i = a_j$ for some j). Then $\phi(a_1, ..., a_m)$ is true in (A_1, R_1) iff $(\psi(b_1, ..., b_k) \wedge \gamma(c_1, ..., c_l))$ is true in (A_1, R_1) iff $(\psi(\pi(b_1, ..., \pi(b_k))) \wedge \gamma(\pi(c_1), ..., \pi(c_l)))$ is true in (A_2, R_2) (by the induction assumption) iff $\phi(\pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2).

The second case, where $\phi(x_1, ..., x_m) = \sim\psi(x_1, ..., x_m)$ (for some formula $\psi(x_1, ..., x_m)$, is similar to the first case.

The third case is when $\phi(x_1, ..., x_m) = (\exists x)\phi(x, x_1, ..., x_m)$. Then $(\exists x)\phi(x, x_1, ..., x_m)$ is true iff $\phi(a, a_1, ..., a_m)$ is true for some $a \in A_1$, iff $\phi(\pi(a), \pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2) iff $(\exists x)\phi(x, \pi(a_1), ..., \pi(a_m))$ is true in (A_2, R_2).

On the other hand, we consider a pure sentence ϕ of degree n. The cases here are nearly identical to the above three cases - we give the case for existential quantification.

In that case, $\phi = (\exists x)\psi(x)$ for some ψ. Then ϕ is true in (A_1, R_1) iff $\psi(a)$ is true in (A_1, R_1) for some $a \in A_1$ iff $\psi(\pi(a))$ is true in (A_2, R_2) (by the induction assumption) iff $(\exists x)\psi(x)$ is true in (A_2, R_2).

Section 1

Theorem 1.1

In the proof, it is remarked that there are only c formulas with constants in S, hence S' has no higher cardinality than S. In the non-denumerable case, this does not at all seem obvious to me, so here is an explanation.

Let F_n be the set of all formulas (with constants in S) of length n. Informally, the set of all sequences of length n (where $n > 0$) made up of only $(,), \exists, \sim, \vee$, and constants in S, has the same cardinality as C^n, where $C = \cup\{\{(\}, \{)\}\}, \{\sim\}, \{\vee\}, \{\exists\}, \}$ (and C has cardinality c). Now, $\overline{\overline{C \times C}} = c$, as $C \times C \cong C$ (by the fundemental theorem of cardinal arithmetic). Obviously the set of all formulas (with constants in S) of length n is not equal to C^n, but it is (informally) a subset of it. So $F_n \preceq C^n \cong c$. But it is obvious that there is a subset of F_n of cardinality c, meaning $c \preceq F_n$, and thus $F_n \cong c$ (by the Bernstein-Schröder Theorem).

Thus we know that, for each i, $\overline{\overline{F_i}} = c$. Now, the set of all formulas with constants in S is equal to $F : \cup\{F_1, F_2, ..., F_n, ...\}$. We can show that $F \cong c$ in the following way. We show that $F \cong \mathbb{N} \times F_j$ (for any j; which one doesn't matter), and because $\mathbb{N} \times F_i \cong c$ (by the law of multiplicative absorpition), this would imply $F \cong c$. Well, we know that each F_i is well ordered (because they have some cardinality at all). Note that F is a union of pairwise disjoint sets. So, for each $x \in F$, let $\phi(x) = \langle i, y\rangle$ where i is the F_i such that $x \in F_i$, and, if x is the αth element of F_i, let y be the αth element of F_j. Because ϕ is a 1-1 correspondence between F and $\mathbb{N} \times F_j$, it follows that $F \cong c$, which is what we set out to show.

The above argument is certainly informal, and not technically correct (for instance, we are using cardinality theorems on things that are not even formally sets). If we wanted to be more formal about it, we could try to represent the connectives (and parentheses) as sets. To do this, consider any set S. We must be able to find some five sets a_0, a_1, a_2, a_3, a_4 outside of S (or else V would be a set). Then we let $(= a_0,) = a_1, \exists = a_2, \sim = a_3$, and $\vee = a_4$, and then the set of all formulas with constants in S would just be a subset of $H \times H$ where $H = \cup\{\{(\}, \{)\}\}, \{\exists\}, \{\sim\}, \{\vee\}, S\}$ (which has cardinality c), and we can proceed in a similar way as we did in the informal argument.

Also, we are using the fundemental theorem of cardinal arithmetic (FTCA), which requires the axiom of choice. Actually, we can alter the proof of the FTCA (in chapter 9) so that $A \cong A \times A$ follows if A is infinite and we only know that that A itself has a choice function. This works as, importantly, Kuratowski's maximal principle (which is used in the proof of Theorem 6.1 of Chapter 9) does not require AC to hold; only that A is closed under chain unions and has a choice function. Throughout the proof of the FTCA (and the preceeding lemmas), whenever we use the fact that AC holds, for whatever set we need to have a choice function, we can show that it is also well ordered.

In particular, the additional assumptions to get the result in the preceeding paragraph would be (the below theorems and lemmas are from chapter 9):
Theorem 6.1: A is well orderable and A' is well orderable.
Theorem 6.2: A and B are well orderable (actually we don't need to know that B is well orderable).
Lemma 6.3: M and M' are well orderable.
Theorem 6.4: A is well orderable.

Exercise 1.1

This is just a matter of redifining truth over A so that a sentence X with constants in A is true over A iff it is true in the relational system (A, R). Then the rest of the proof is the same.

Exercise 1.2

Assuming the axiom of choice, this means that every infinite set has a denumerable subset. Now, recall that the Skolem-Löwenheim theorem states that if ϕ (having no constants) is satisfiable in any infinite relational system at all, then it is satisfiable in a denumerable relational system. Well, suppose ϕ is satisfiable in some infinite relational system (A, R). Then A can be well ordered (as the axiom of choice is assumed). Suppose that ϕ has n free variables, meaning it can be written as $\phi(x_1, ..., x_n)$. Because it is satisfiable in (A, R), $\phi(a_1, ..., a_n)$ is true in (A, R) for some $a_1, ..., a_n$ in A. Then, by

the axiom of choice, there is a denumerable subset A_0 of A which includes all of $a_1, ..., a_n$. By Exercise 1.1, there is some denumerable subset B of A such that B reflects A, and B is an extension of A_0. This means that, because $\phi(a_1, ..., a_n)$ is true in (A, R), $\phi(a_1, ..., a_n)$ will be true in (B, R). Thus ϕ is satisfiable in the denumerable domain B.